Margeaux Corrigan

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EDUCATION

Brown University, *Sc.B. Electrical Engineering*, 4.0/4.0 GPA — Providence, RI | **Expected Graduation December 2029**

Relevant Coursework: Data Structures & Algorithms, Statics & Dynamics, Ordinary Differential Equations, Linear Algebra

In Progress: Fundamentals of Computer Systems, Electricity & Magnetism, Statistics, PDEs | Spring 2027: Electrical Circuits & Signals

TECHNICAL SKILLS

Circuit Design & Hardware: Schematic capture & PCB Layout (KiCad), component selection, wiring diagrams, soldering

Embedded Systems: bare-metal firmware, register-level programming, device drivers, Arm Cortex-M0+/M4F, I2C, SPI, UART

Debugging & Test: oscilloscope, logic analyzer, NanoVNA, multimeter, bench power supply, OpenOCD, SWD, Unity/CTest, pytest

Languages: C, Python, Ada, MATLAB, Java

Software & Tools: Git/GitHub, CMake, GitHub Actions CI, Linux/bash, arm-none-eabi-gcc, NumPy, pandas, Matplotlib

Mechanical & CAD: SolidWorks, Fusion 360, FEA (SolidWorks), 3D printing, laser cutter

EXPERIENCE

GE Aerospace, *Incoming Avionics Systems Engineering Co-op* — Fall 2027

- Earned co-op offer via interview at GE Aerospace LIFT Summit, selected as 1 of 61 from 3,800+ applicants to attend

Brown University Interactive 3D Vision & Learning Lab, *Undergraduate Researcher* — Providence, RI | Jun 2026 – Present

Advisor: Srinath Sridhar, Assistant Professor | Brown University

- Co-maintained a 20-unit camera rig fleet: ran status checks, repaired 6+ faulty units, debugged multi-camera start/stop faults
- Assisted with schematic and KiCad PCB layout for a BJT-driven LED module slated for the next camera rig revision
- Built egocentric capture unit for partner lab: selected connectors and cabling, designed 3D-printed enclosure, installed hardware
- Co-leading data capture project for the partner lab; gather their data requirements and report progress at weekly meetings

Persistence Labs Inc., *Strategy Intern* — Remote | Jun 2024 – Oct 2024

- Ran beta testing across 10+ iterations for AI social media app; presented findings to founder, informing feature prioritization

ENGINEERING PROJECTS

Bare-Metal Driver Stack, *Independent Project* — RP2040, STM32G4, C | May 2026 – Present

- Wrote register-level C drivers for RP2040 GPIO, SysTick, UART, I2C, and BMP390, no HAL; verified with a logic analyzer
- Built a Unity/CTest host suite covering every BMP390 function via link-time fakes; injected NAK and timeout faults in CI
- Hand-wrote STM32G431RB startup: linker script, vector table, FPU enable, reset handler; boot verified over SWD
- Traced failed init to a BMP390 power-on value contradicting the datasheet; set drivers and tests to share one verified constant

Rocket Payload Sensor System, *Sole Developer* — RP2040, Ada | Feb 2026 – Apr 2026

- Wrote bare-metal Ada firmware under weekly AdaCore engineer review; set BMP390 and LSM9DS1 registers from datasheets
- Deployed on a live high-power launch; logged 377 m AGL apogee, boost past the ±16 G linear range, and descent to recovery
- Built W25Q128 SPI flash logger; sized sample rate to 16 MB and logged 438,858 samples over 25 h with zero dropped intervals
- Found a 1.5x count-to-g conversion error in Python post-flight analysis by checking logged data against static 1 G on pad
- Replaced HAL SPI transmit with register-level transfers after early chip-select release silently truncated flash writes

LEADERSHIP & ACTIVITIES

Brown Rocketry Club, *Avionics Team Lead (2025), Co-President (2026)* — Providence, RI | Sep 2025 – Present

- Co-led weekly meetings and avionics workshops for freshmen new to microcontrollers, building toward a fall launch
- Designed IREC avionics wiring diagrams; split power into five isolated buses for redundancy and competition rules
- Ran SolidWorks FEA on the avionics bay aluminum ring to cut mass while holding margin under expected flight loads
- Delivered the Ada payload that fulfilled a $1K AdaCore sponsorship; arranged the engineer's mentorship for club members

Brown Space Engineering, *Communications Team Member* — Providence, RI | Mar 2026 – Present

- Laid out CubeSat antenna board in KiCad from existing dimensions; calibrated a NanoVNA for antenna characterization
- Stenciled solder paste and hand-placed SMD components on the CubeSat comms board in a clean room for reflow soldering
- Sized UHF Nitinol dipole and S-band patch antennas to CubeSat constraints; corrected patch length for fringing fields

LICENSES, AWARDS, & GRANTS

Amateur Radio Technician License, 2026 | **Brown UTRA Research Award,** 2026 | **Taco Bell Ambition Accelerator Grant,** 2024